

Programming: Analysing Ocean Drifters

Adam M. Sykulski

Associate Professor in Statistics

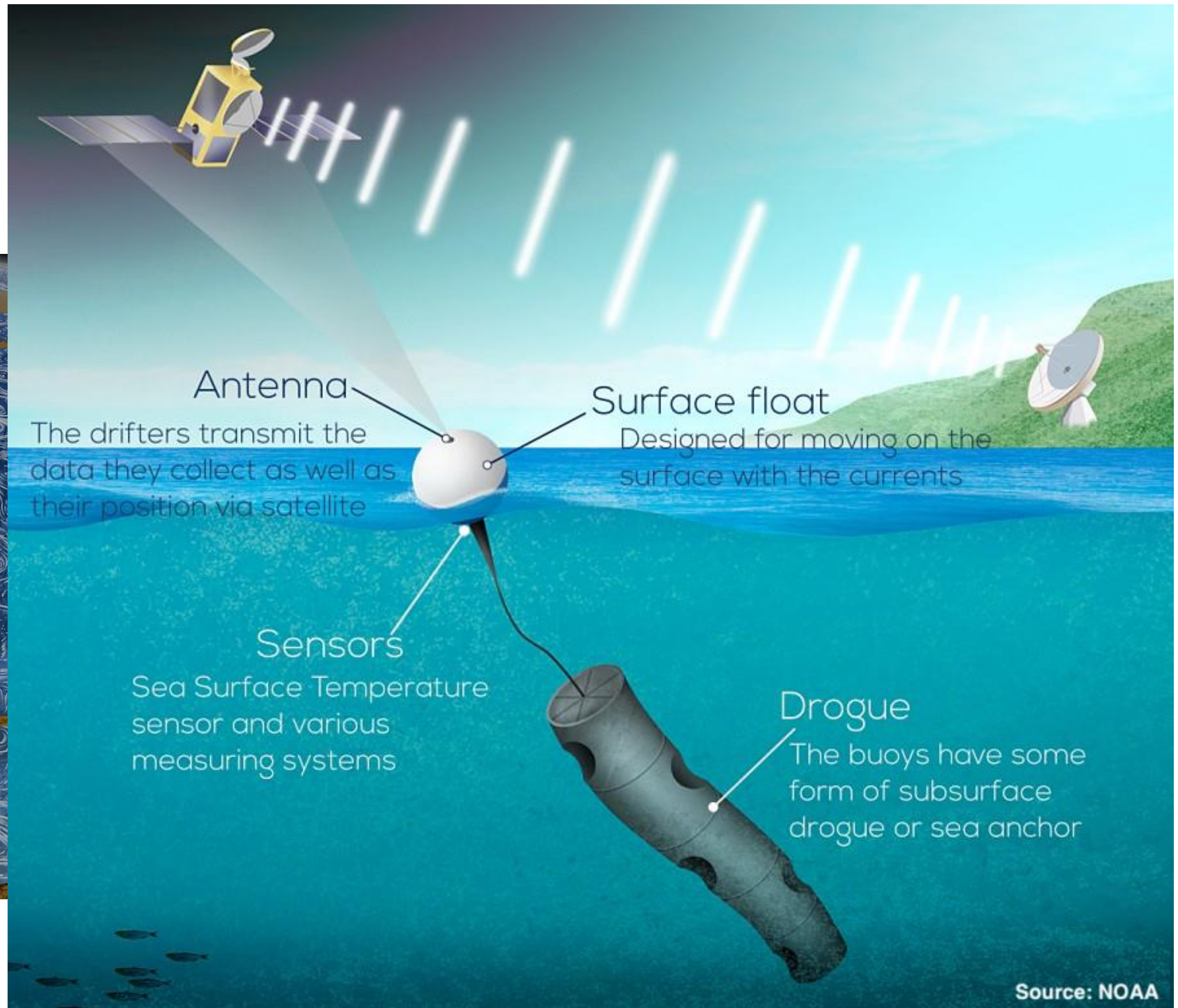
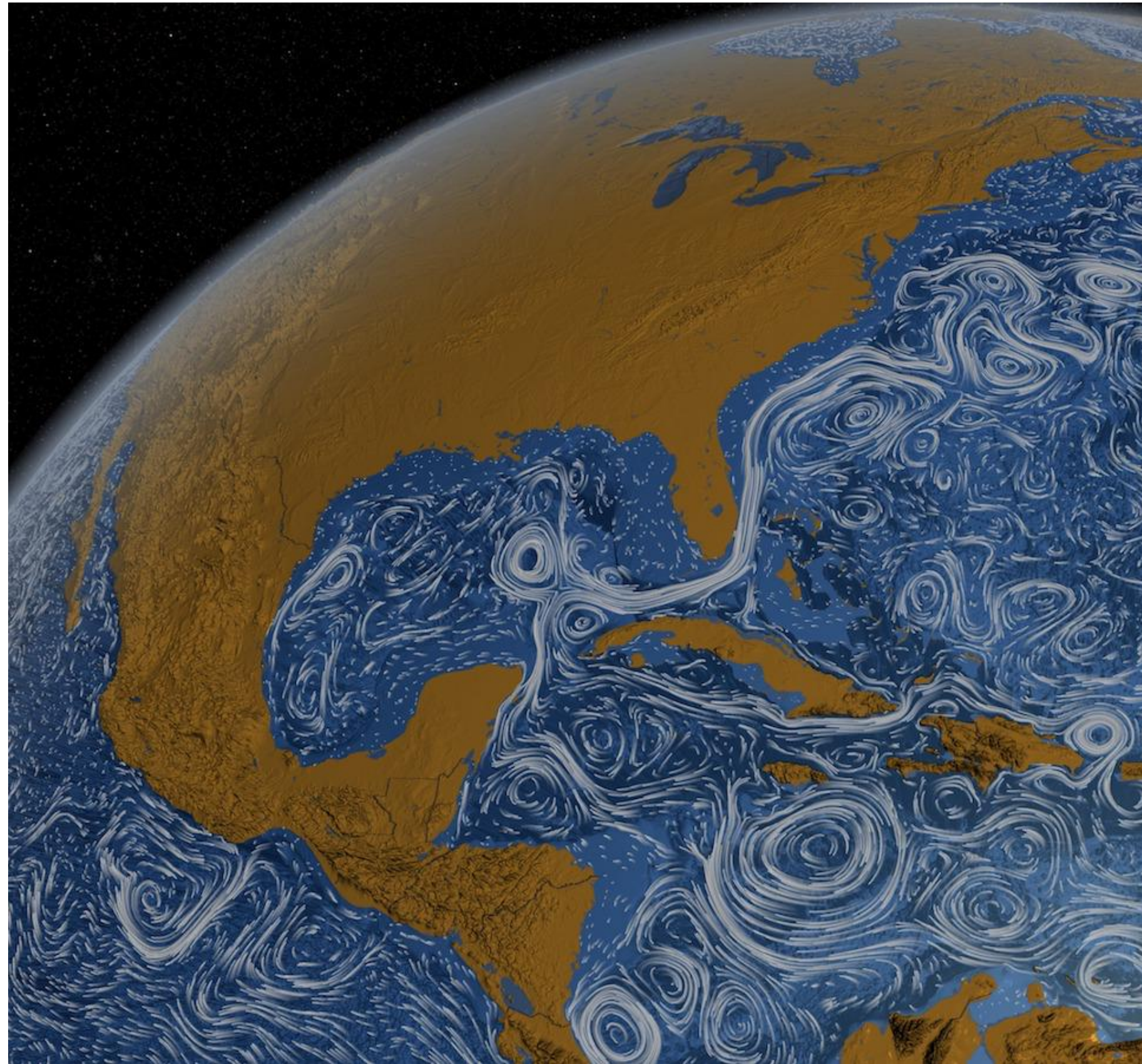
Imperial College London

adam.sykulski@imperial.ac.uk

The goal(s) of this session

- Learning about ocean drifter data, what it is and how to model and analyse it
- Learning how to do statistical analysis and visualisations in R (or Python)
- Bonus: working in small groups, learning to communicate, code efficiently and use resources!

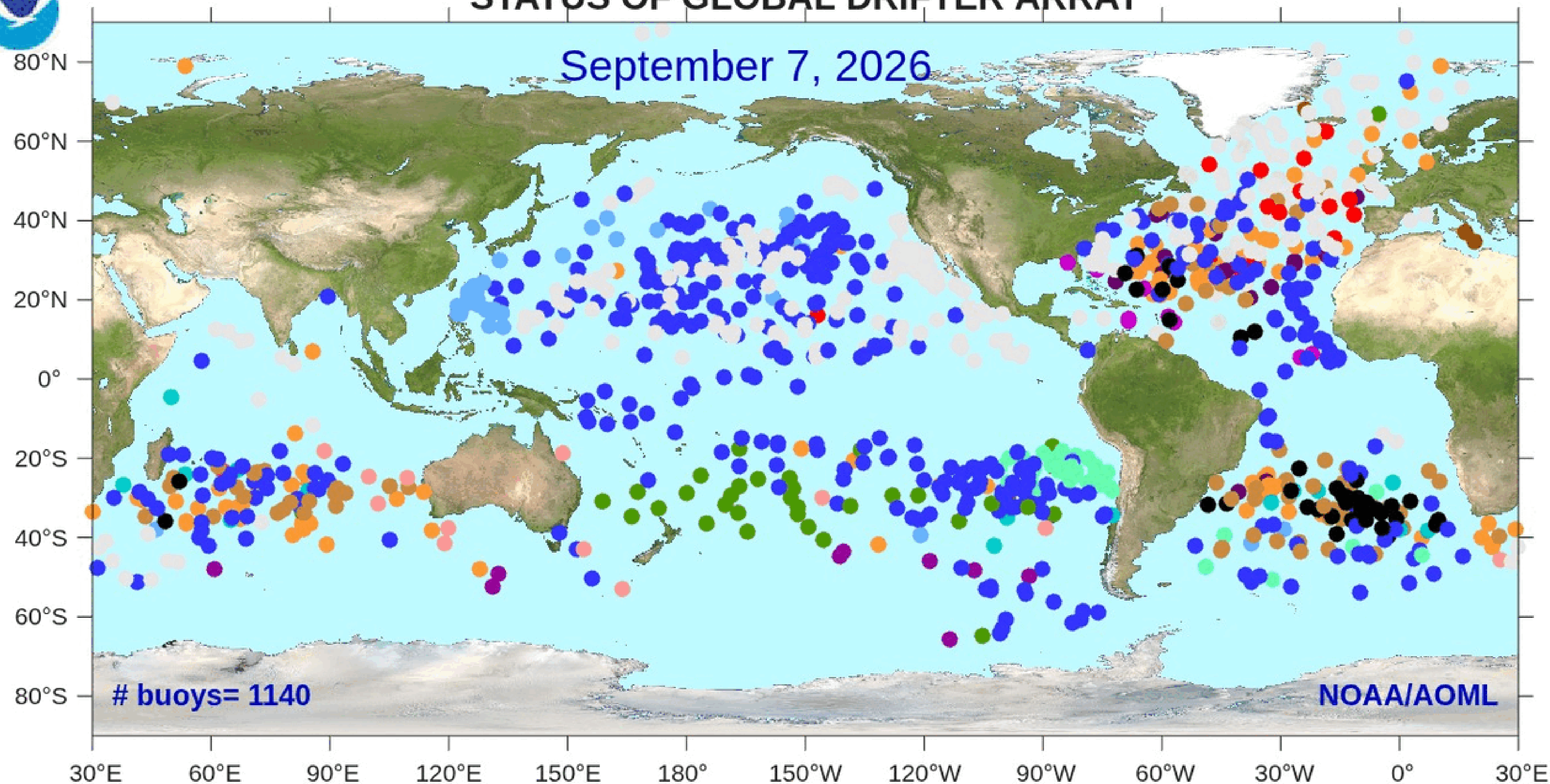
Monitoring our Oceans





STATUS OF GLOBAL DRIFTER ARRAY

September 7, 2026



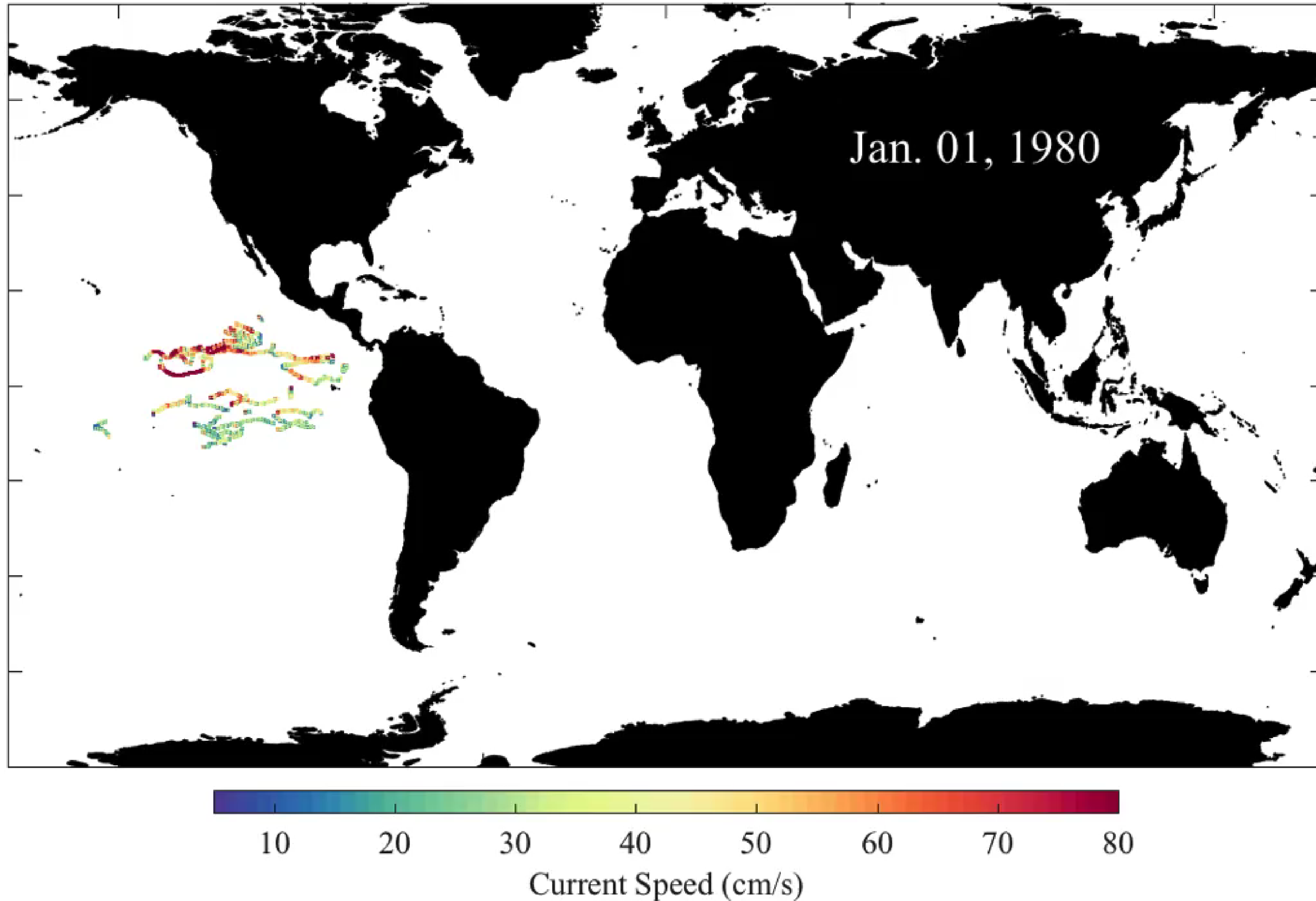
Australia (12)
Brazil (9)
Canada (18)
Chile (38)
France (109)

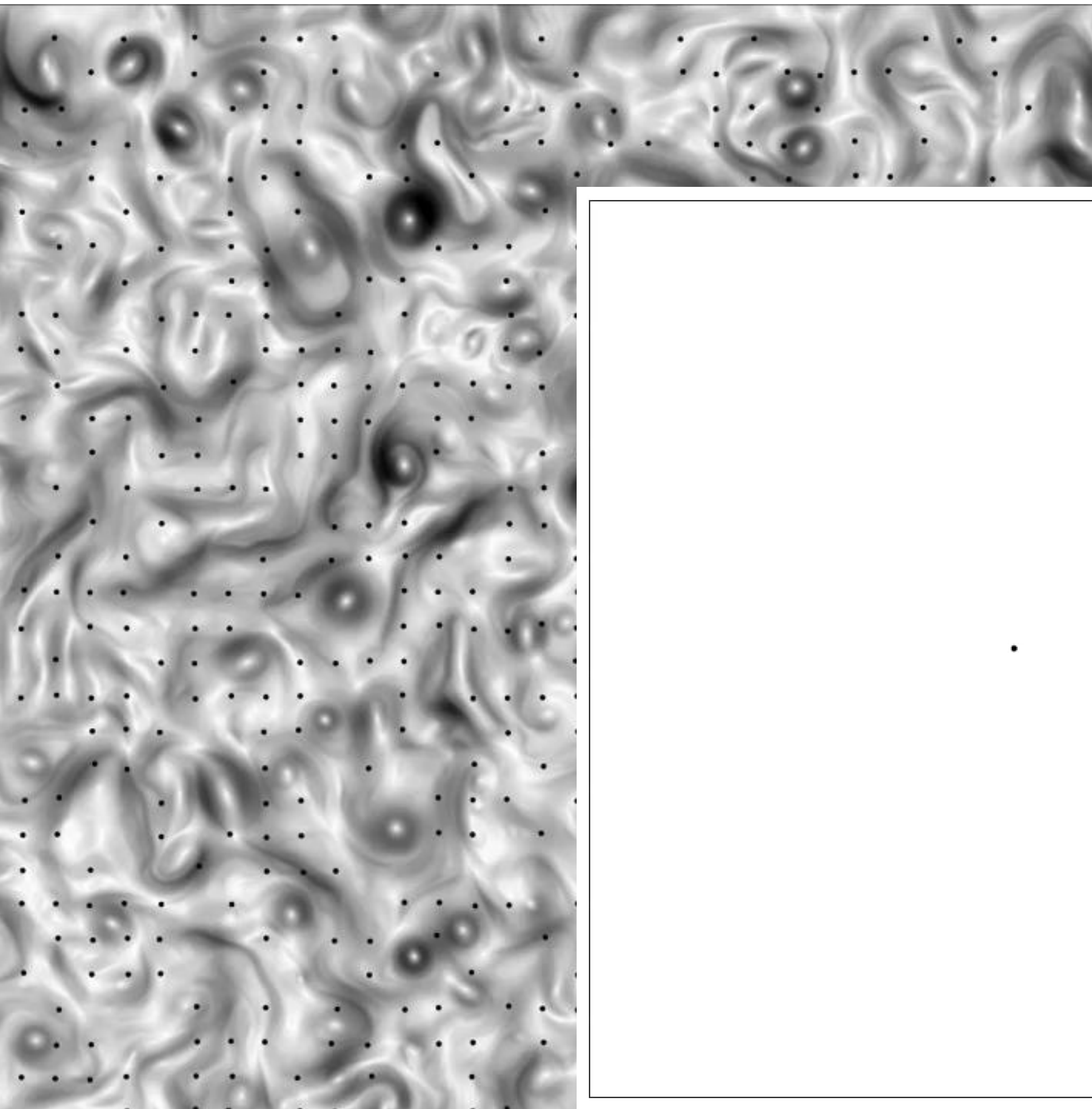
Germany (42)
Iceland (1)
Italy (4)
Japan (10)
Korea, Rep. of (64)

New Zealand (33)
Netherlands (10)
Portugal (18)
Seychelles (1)
South Africa (16)

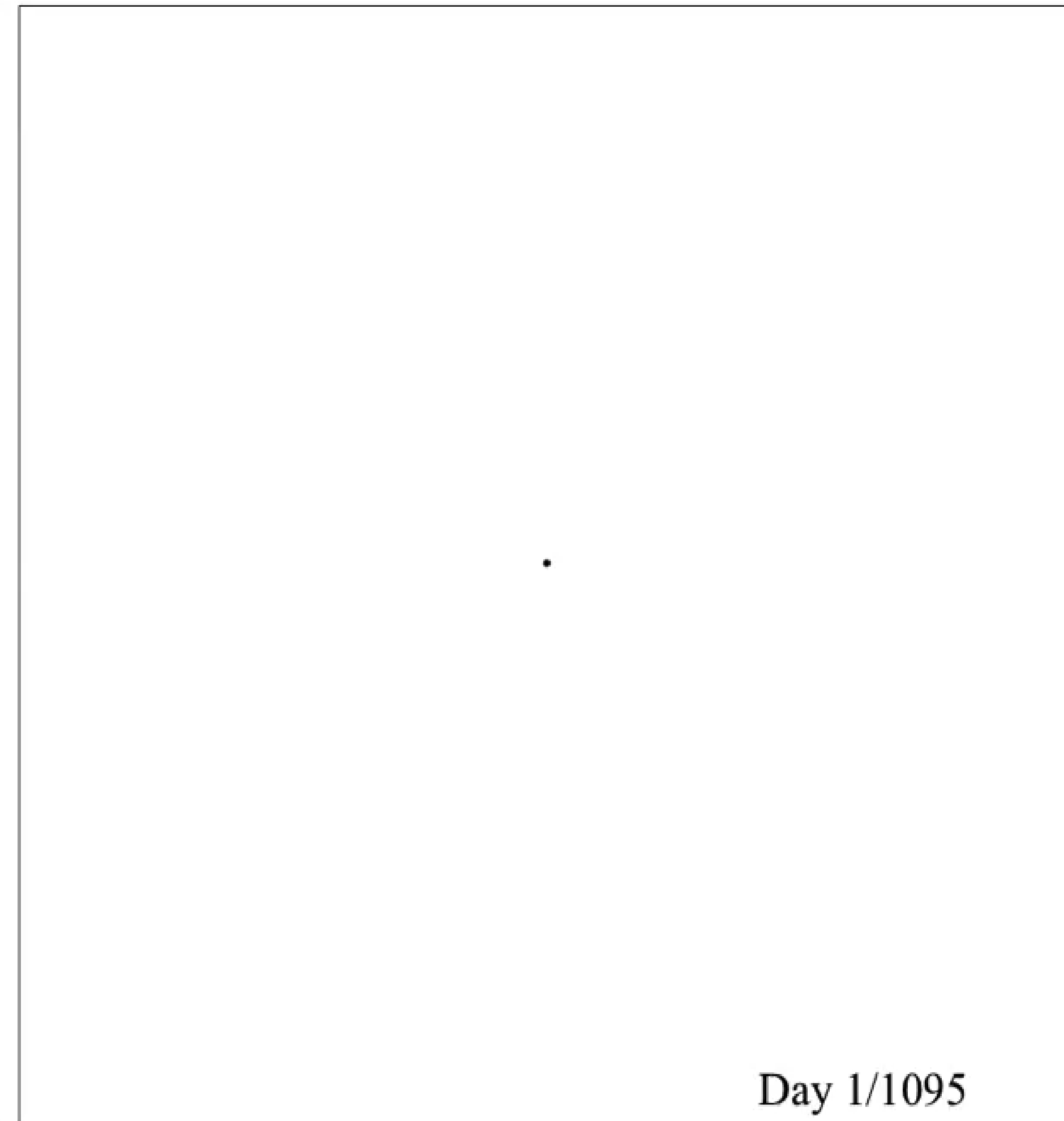
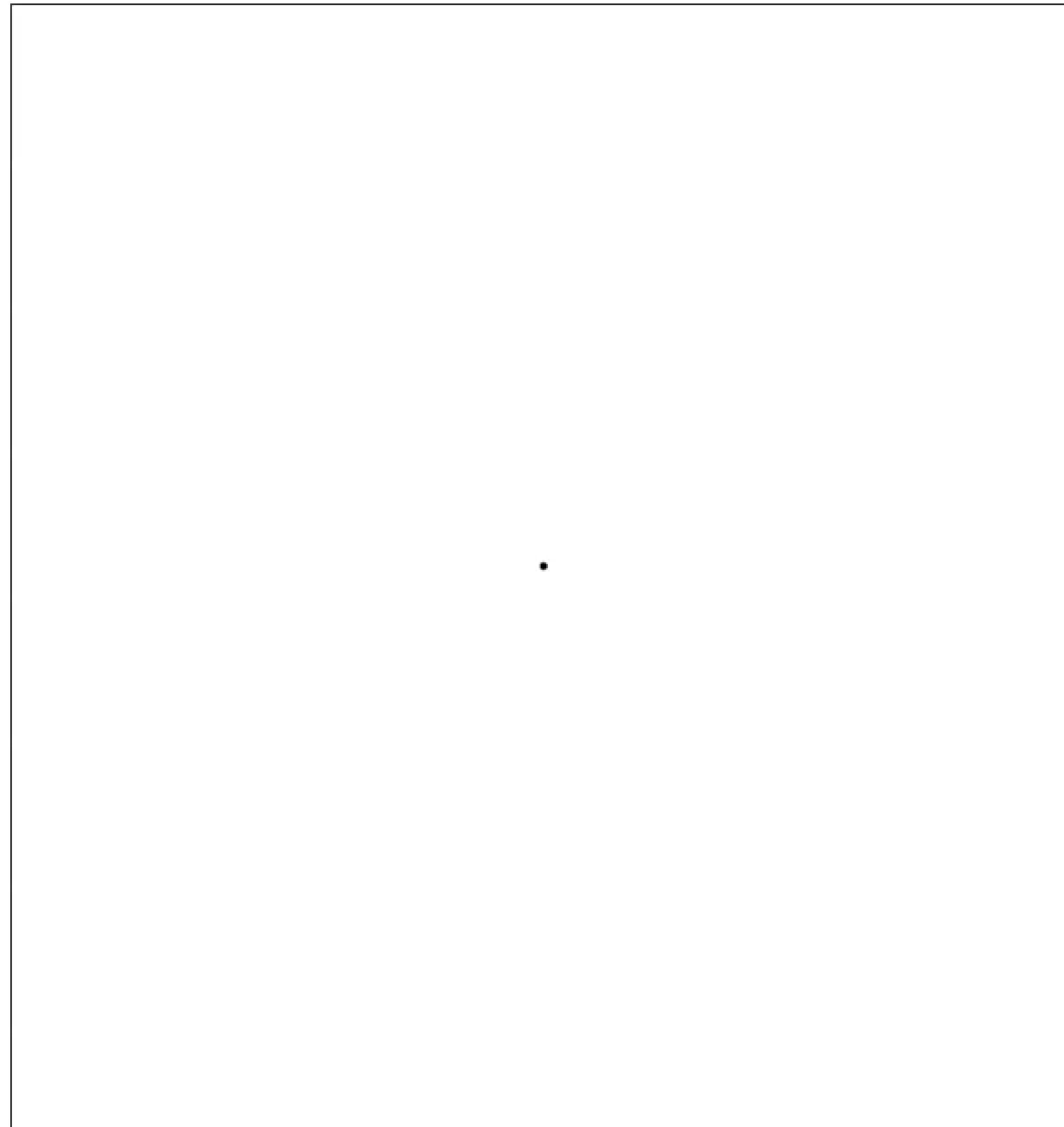
UK (72)
USA (411)
Unknown (272)

Global Drifter Program

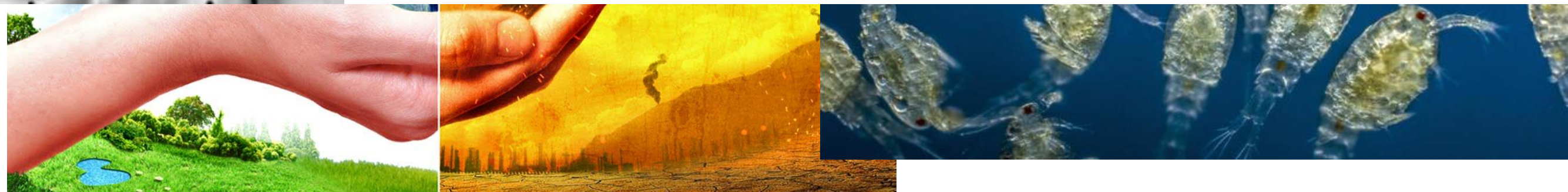




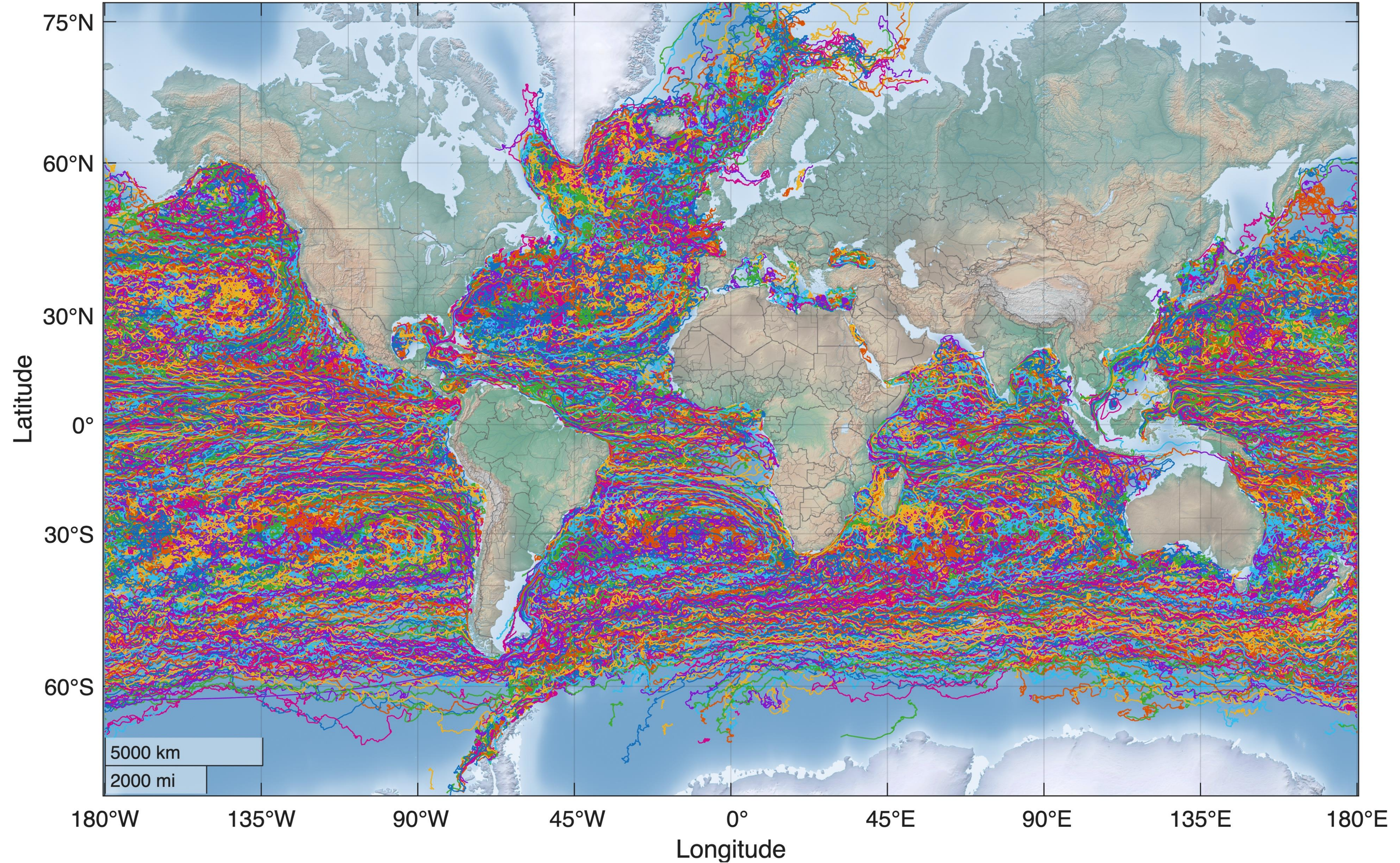
what is this good for?



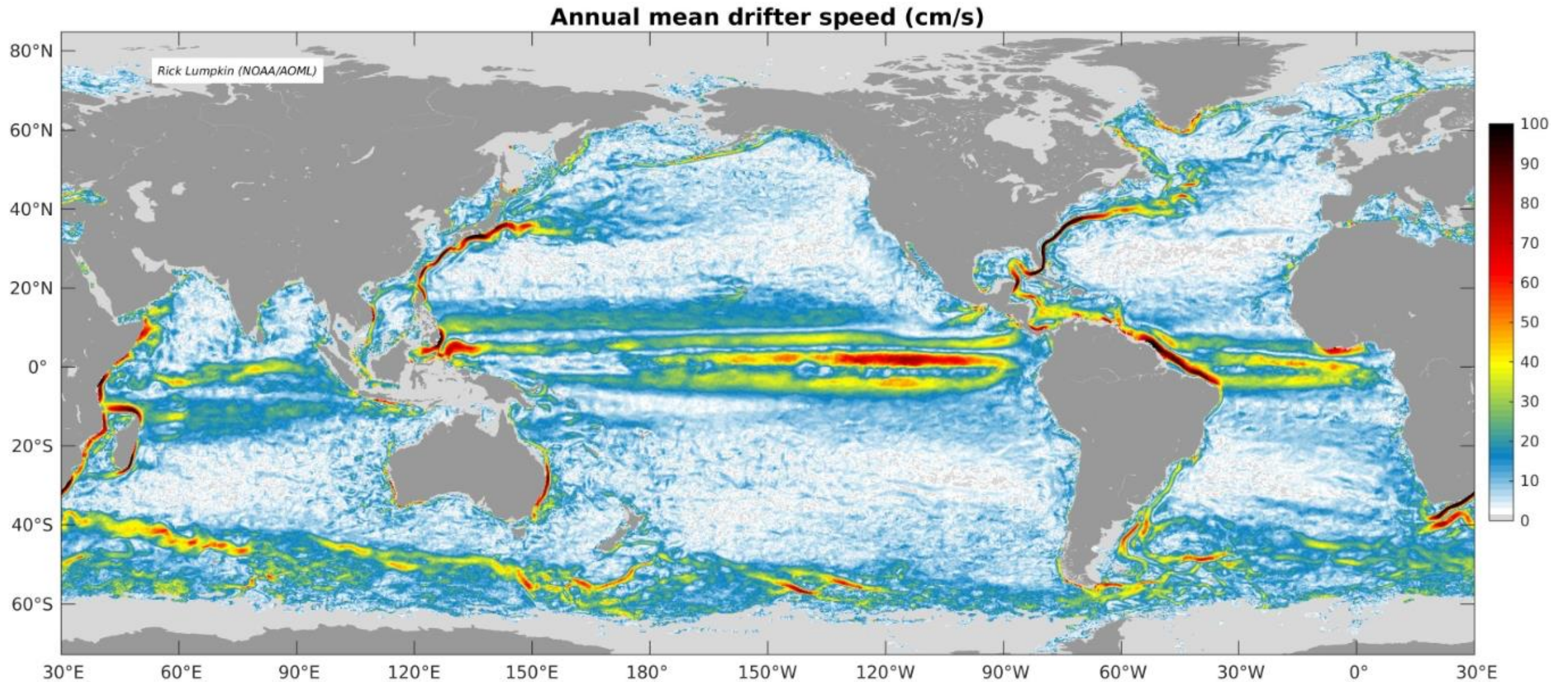
Day 1/1095



Spaghetti Plot – “data dump”

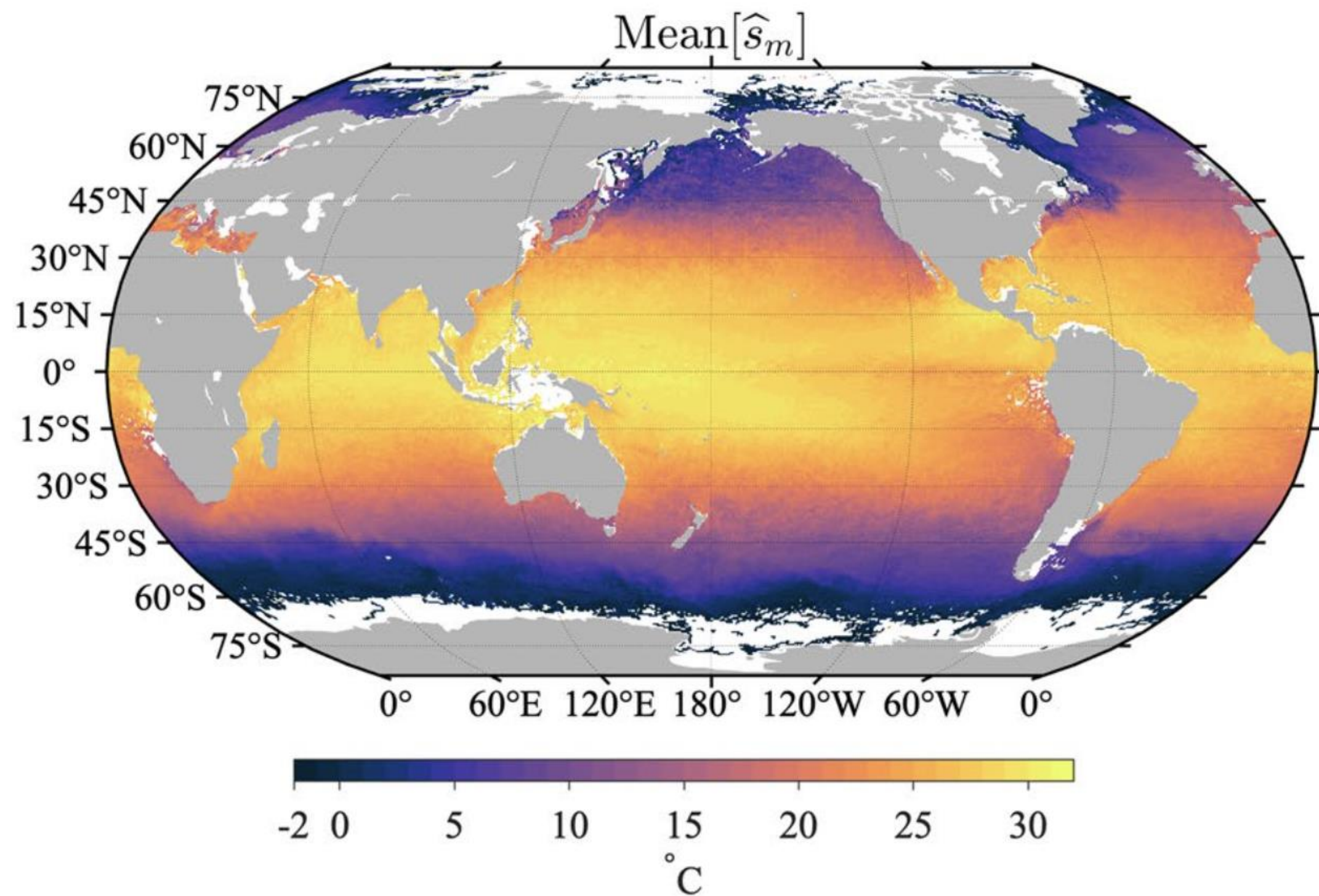


First step: Averaging the data

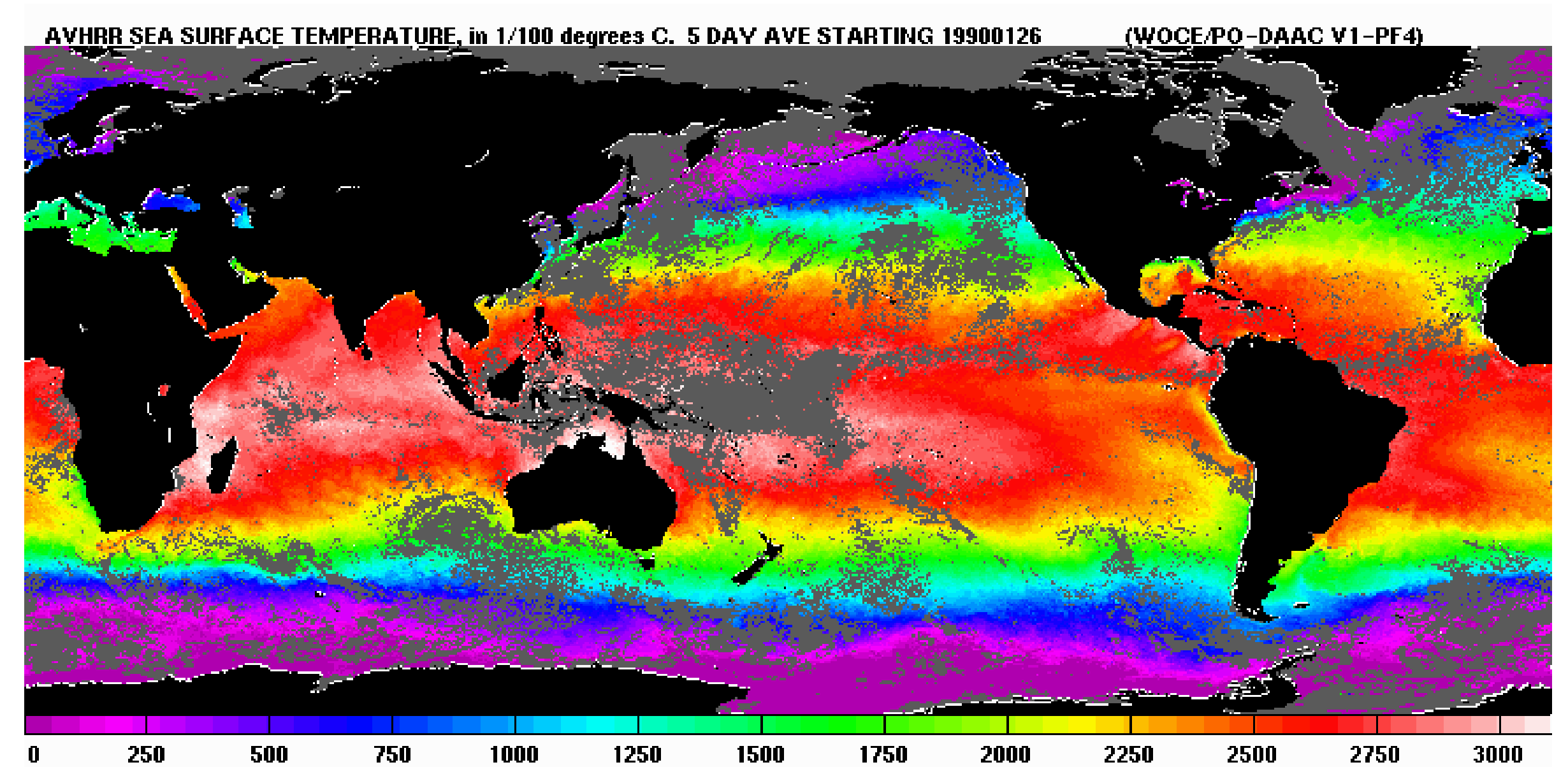


Activity 1: Drifters vs Satellites

Sea Surface Temperature



Drifter SST



Satellite SST

Download data and R scripts from:
github.com/AdamSykulski/OceanDrifterDataInR.git

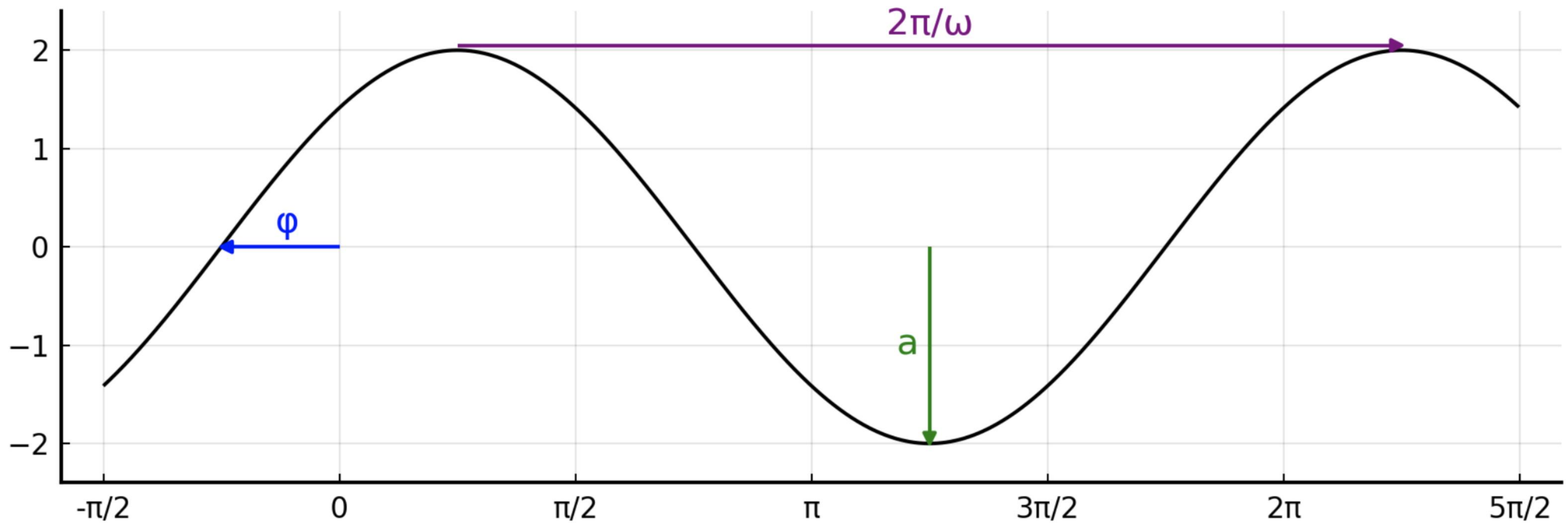
And then head over to RStudio for Activity 1!

Activity 2: Spectral Analysis of Time Series

Consider a basic sinusoid:

$$\eta(t) = a \sin(\omega t + \varphi)$$

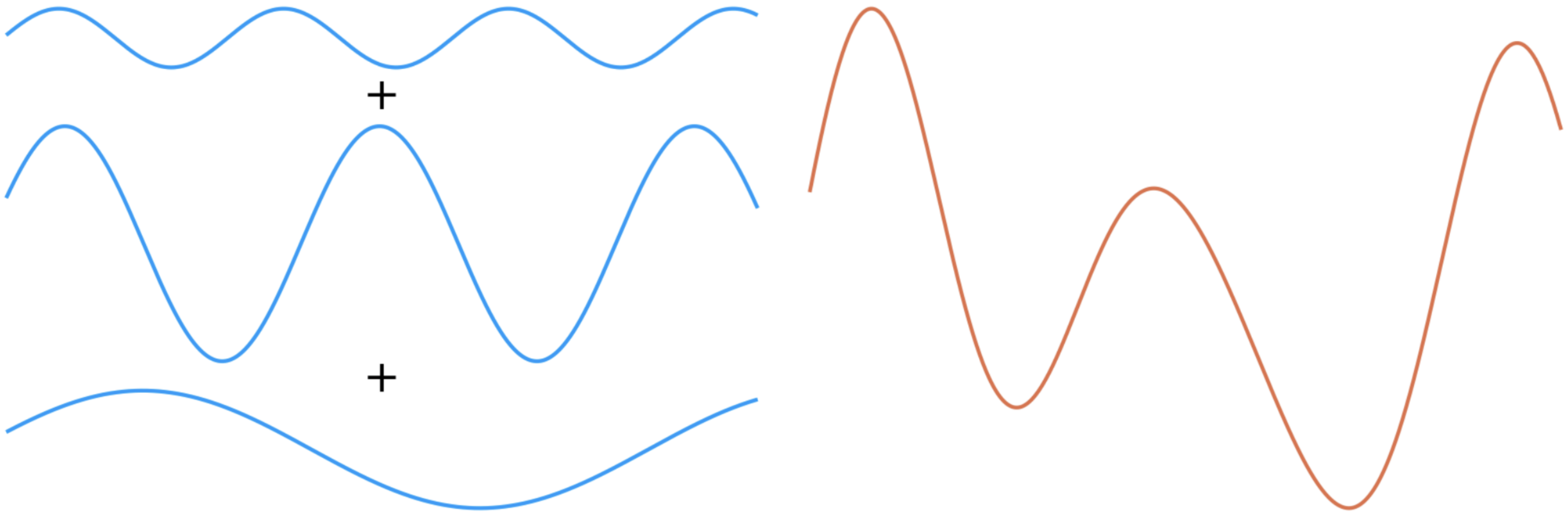
- a is the *amplitude*.
- φ is the *phase*.
- ω is the *angular frequency*.



We can represent more complicated functions as a sum of sinusoids:

$$\eta(t) = \sum_{\omega} a(\omega) \sin(\omega t + \varphi(\omega))$$

with different amplitudes and phases for each component

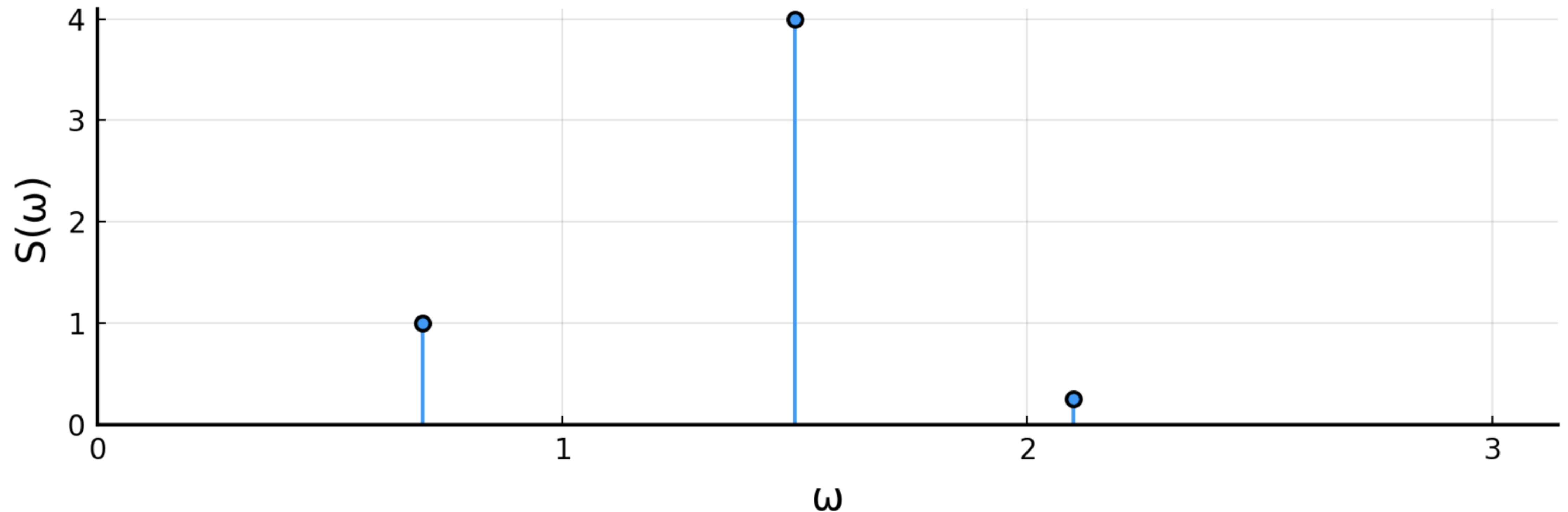


The power spectrum: discrete frequency

The power spectrum is then defined as

$$S(\omega) = |a(\omega)|^2$$

In our example:



We could keep going ... but what if our time series process lives at all frequencies and is stochastic?

This is the field of spectral analysis!

Estimating the power spectral density

Theory: $S_x(\omega) = \lim_{T \rightarrow \infty} \mathbb{E} \left(\frac{1}{2T} \left| \int_{-T}^T x(t) e^{-i\omega t} dt \right|^2 \right)$

Practice: Observe some sample $X_1, \dots, X_N$ at intervals Δ such that

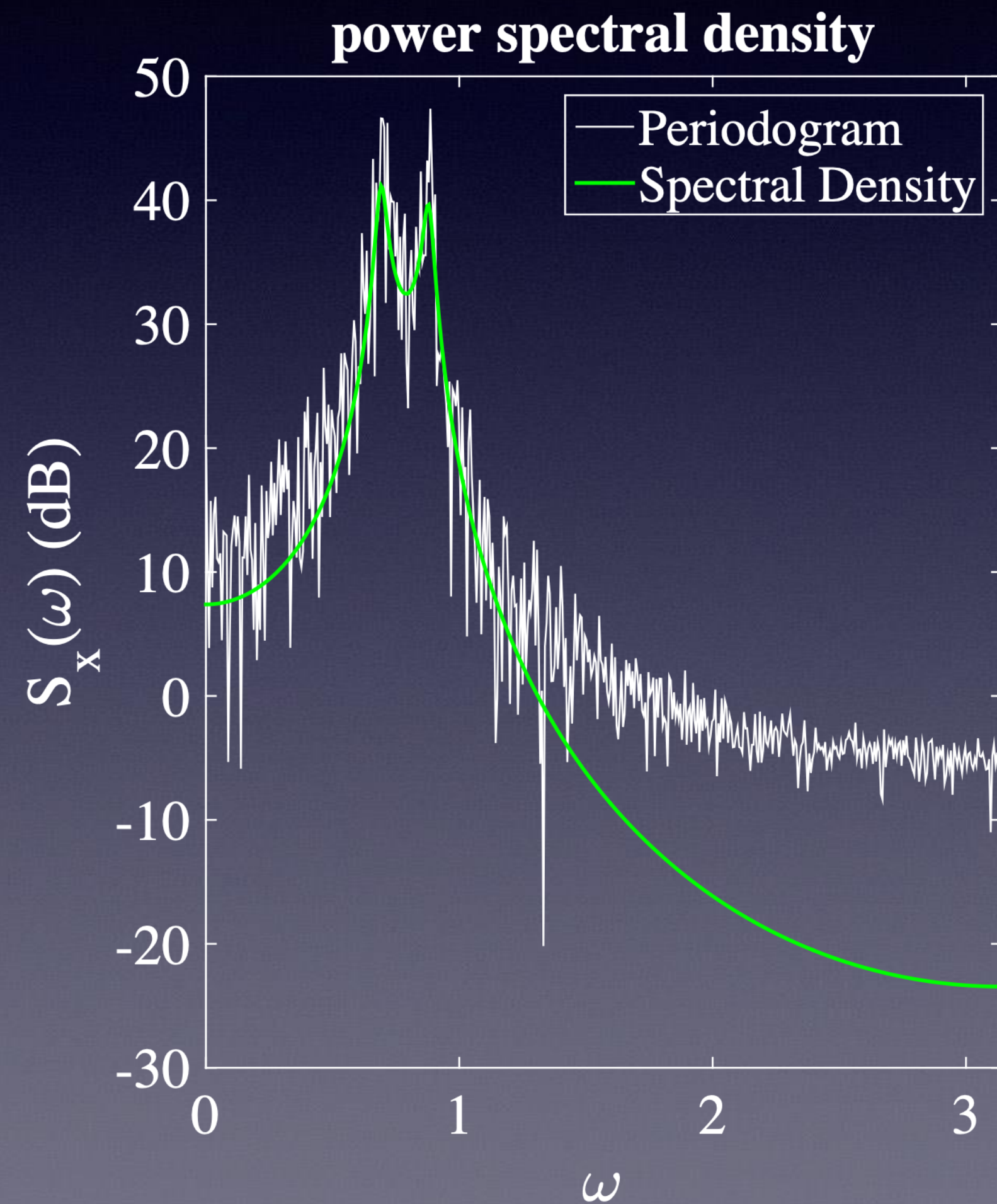
$$\hat{S}(\omega) = \frac{\Delta}{N} \left| \sum_{t=1}^N X_t e^{-i\omega t} \right|^2$$

This is called the *periodogram* and is defined for $\omega \in [-\pi/\Delta, \pi/\Delta]$ where π/Δ is the *Nyquist frequency*.

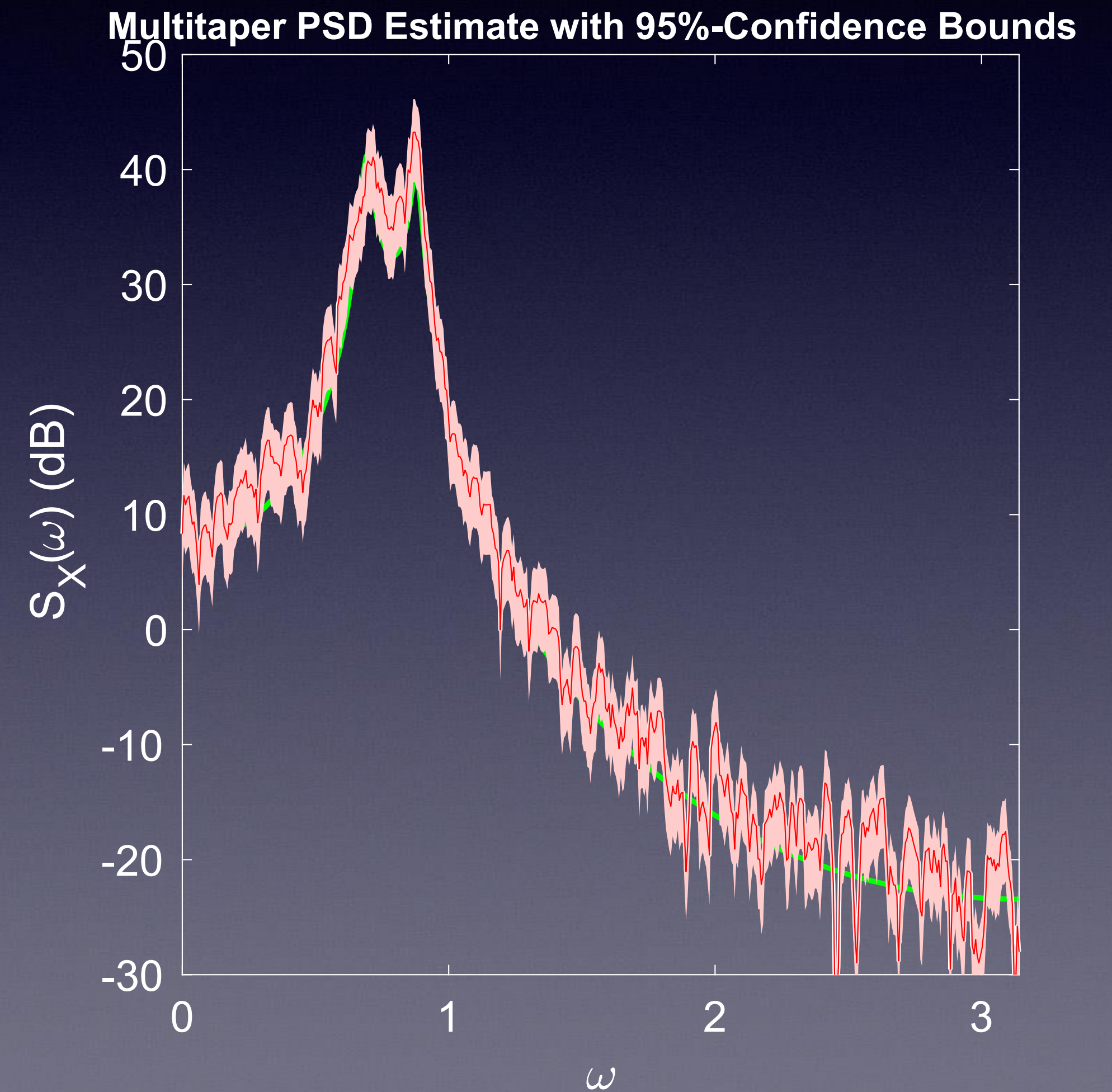
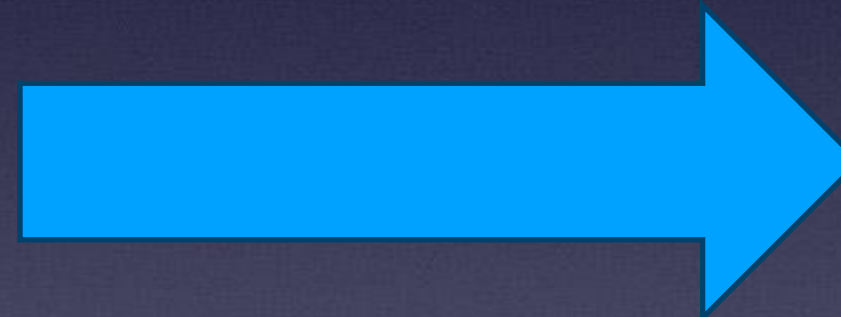
AR(4)

$$x_t = 2.7607x_{t-1} - 3.8106x_{t-2} + 2.6535x_{t-3} - 0.9238x_{t-4} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1)$$

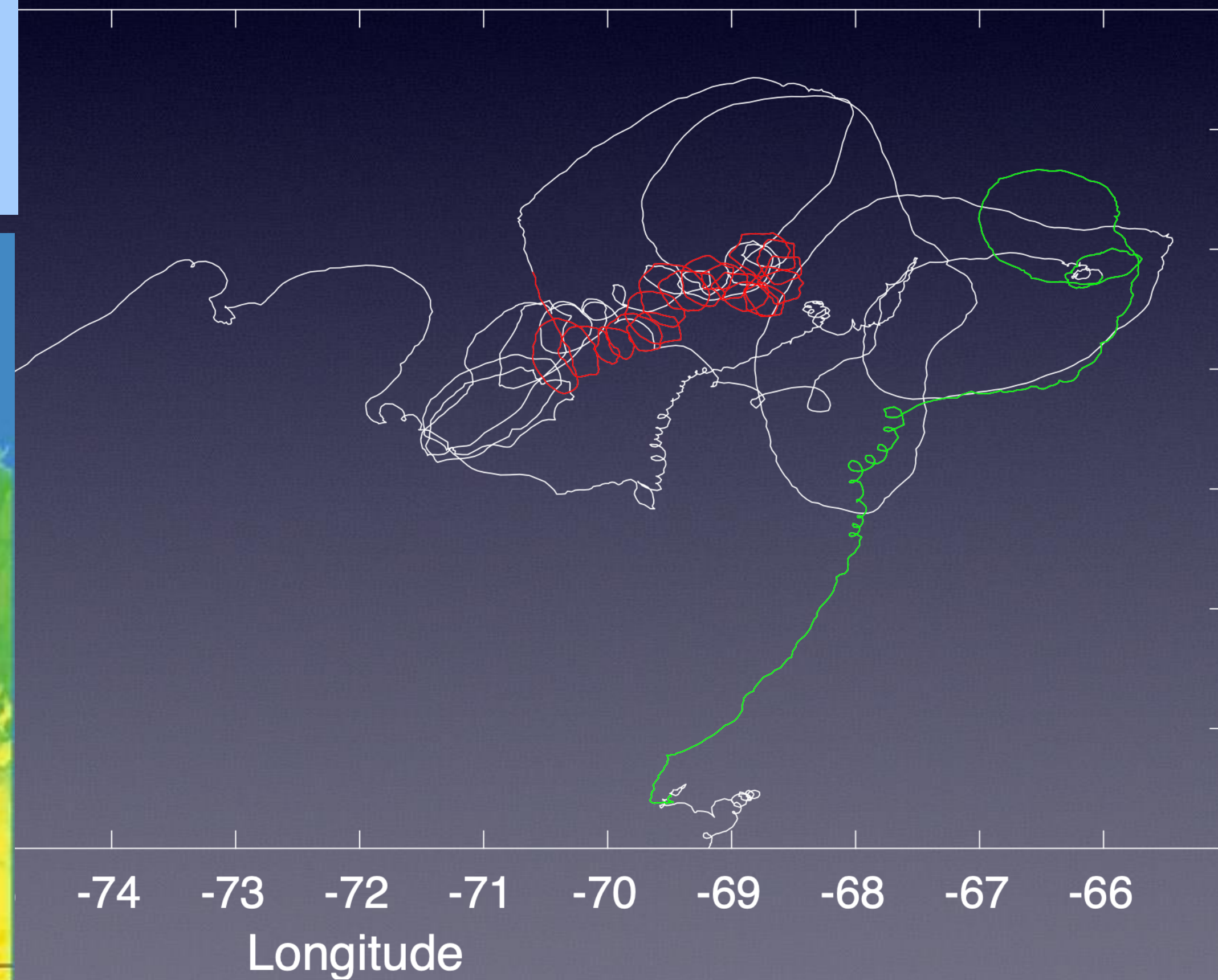
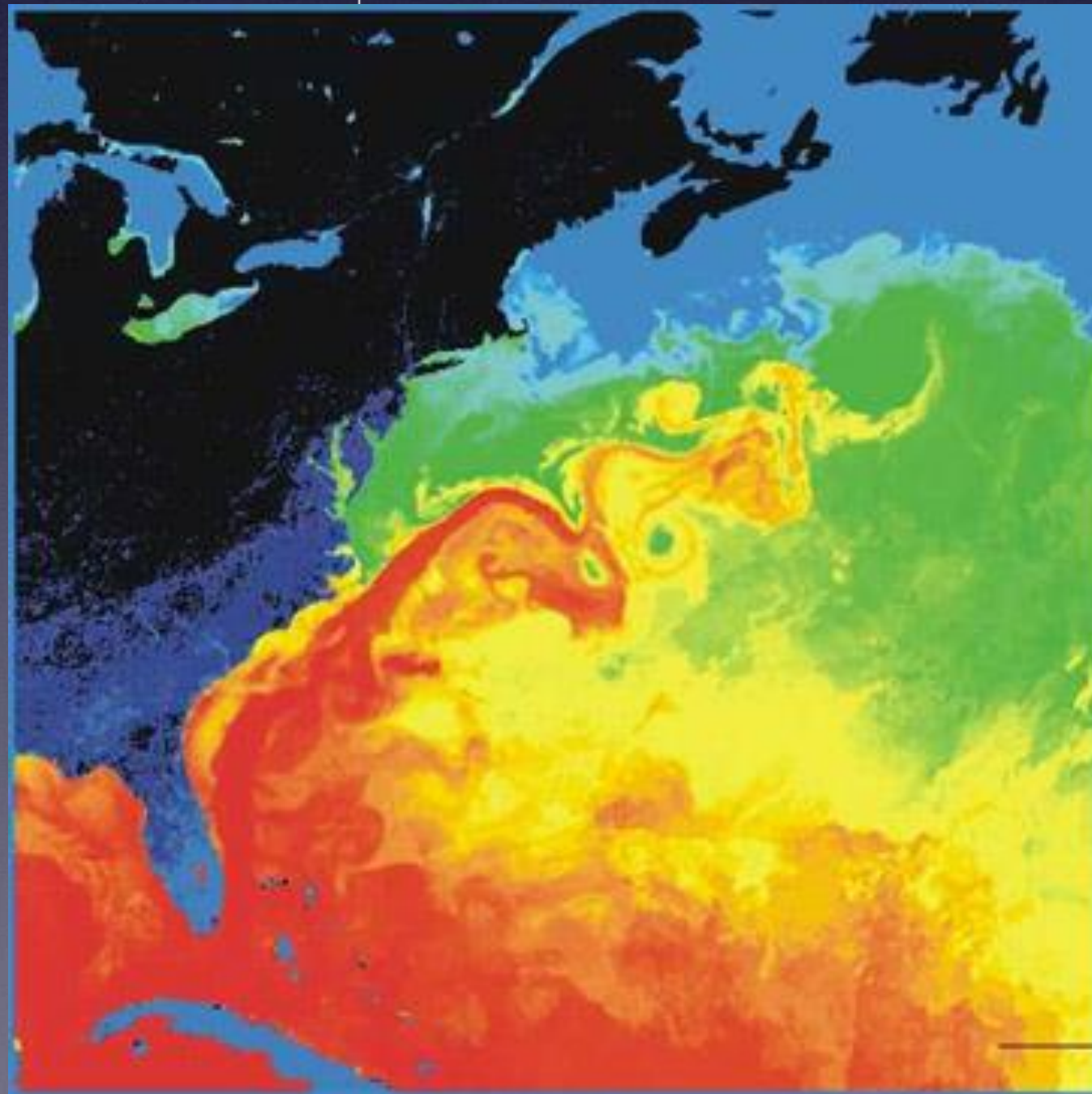
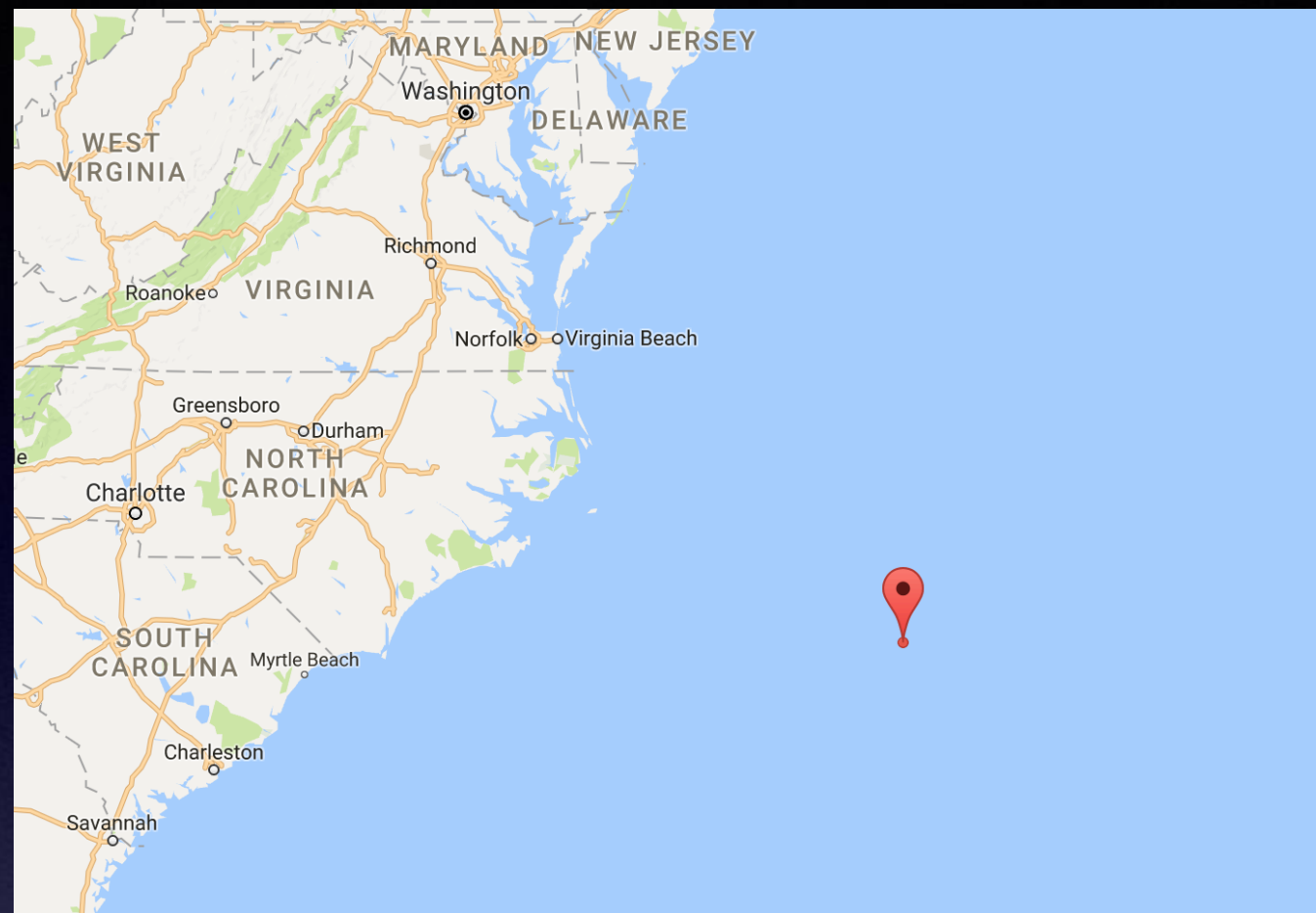
$$S_x(\omega) = \frac{\sigma_\varepsilon^2}{|1 - \sum_{k=1}^4 \phi_k e^{-ik\omega}|^2}$$



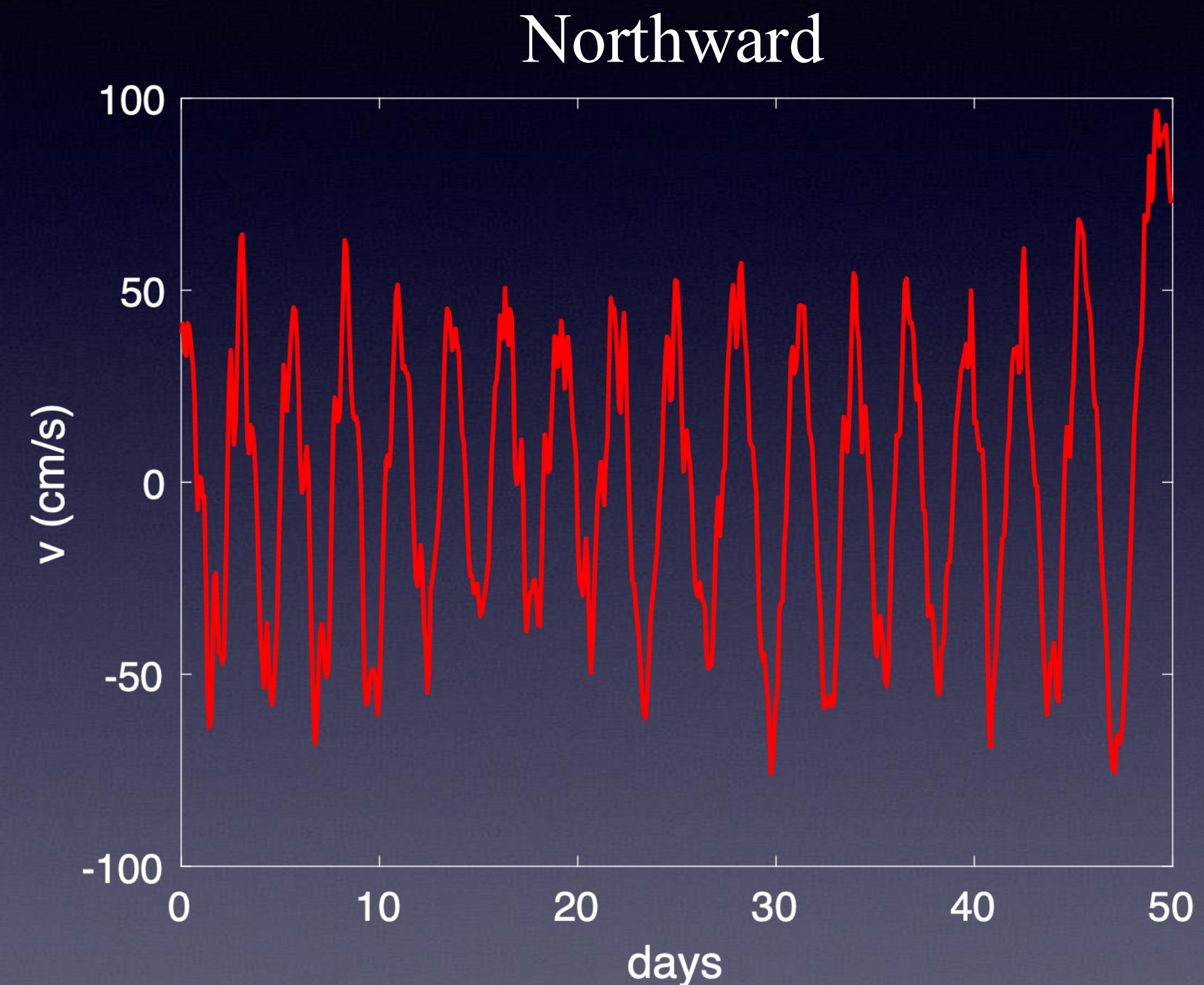
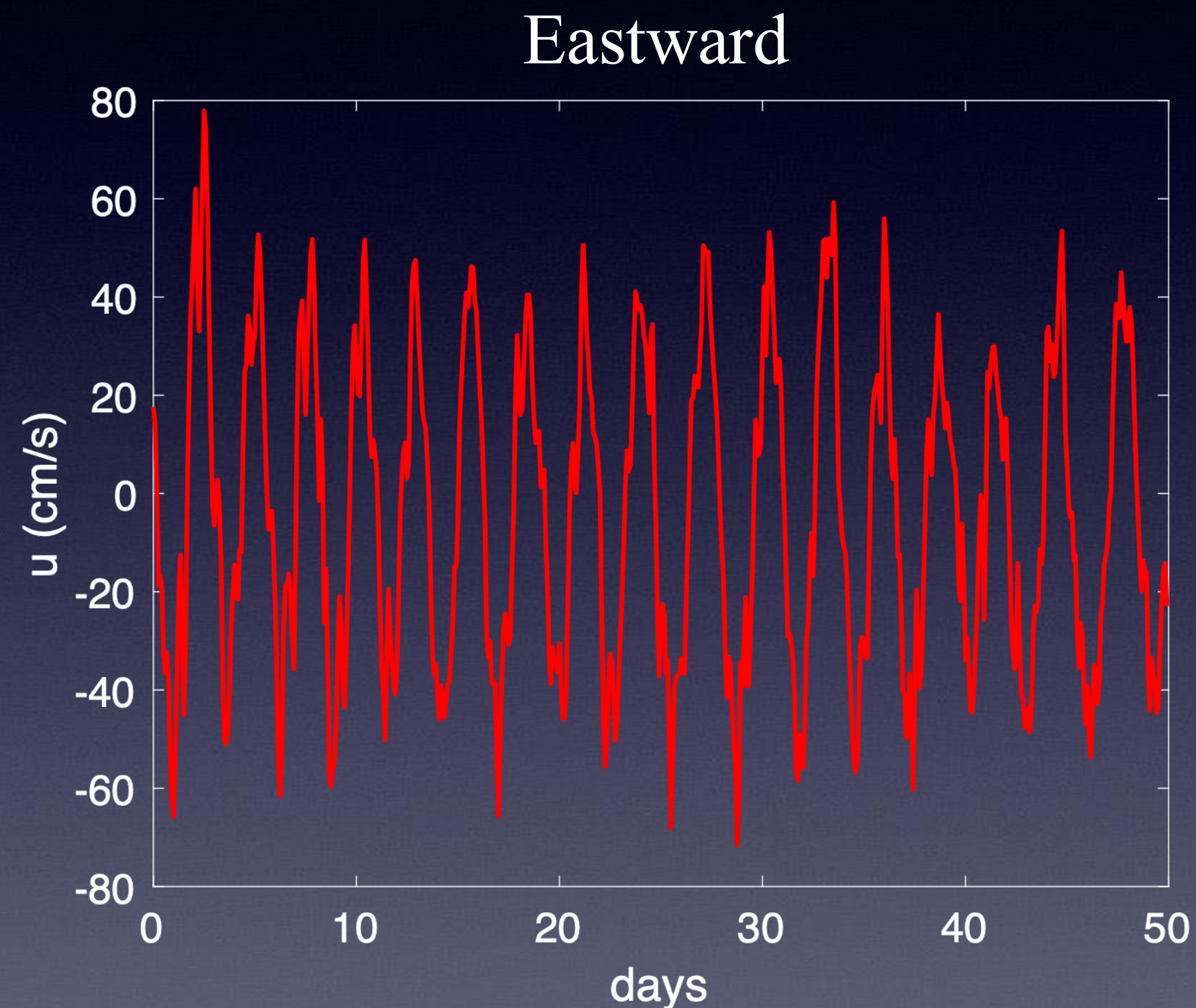
tapering



Oceanographic Drifter Data



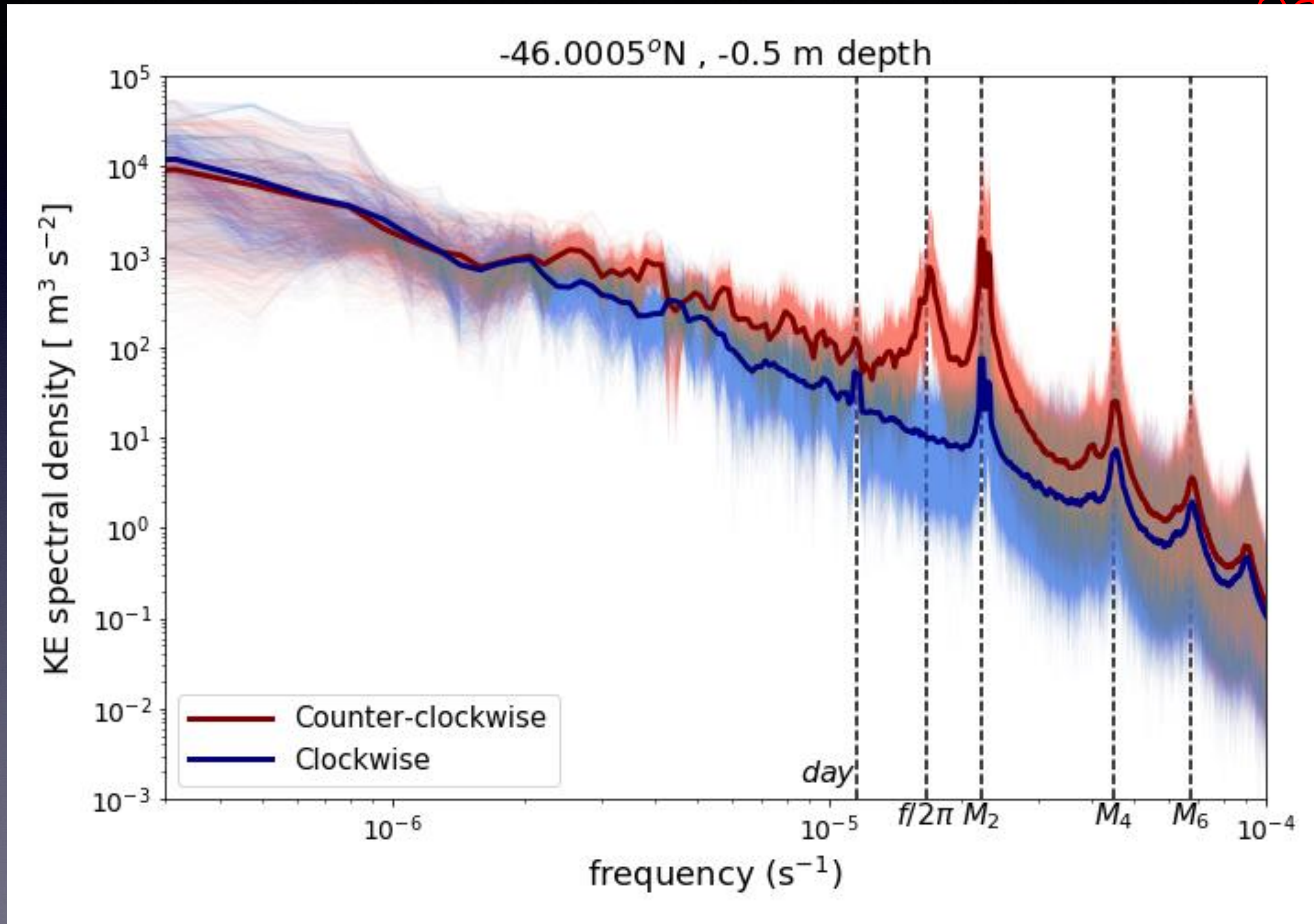
Drifter velocities for:



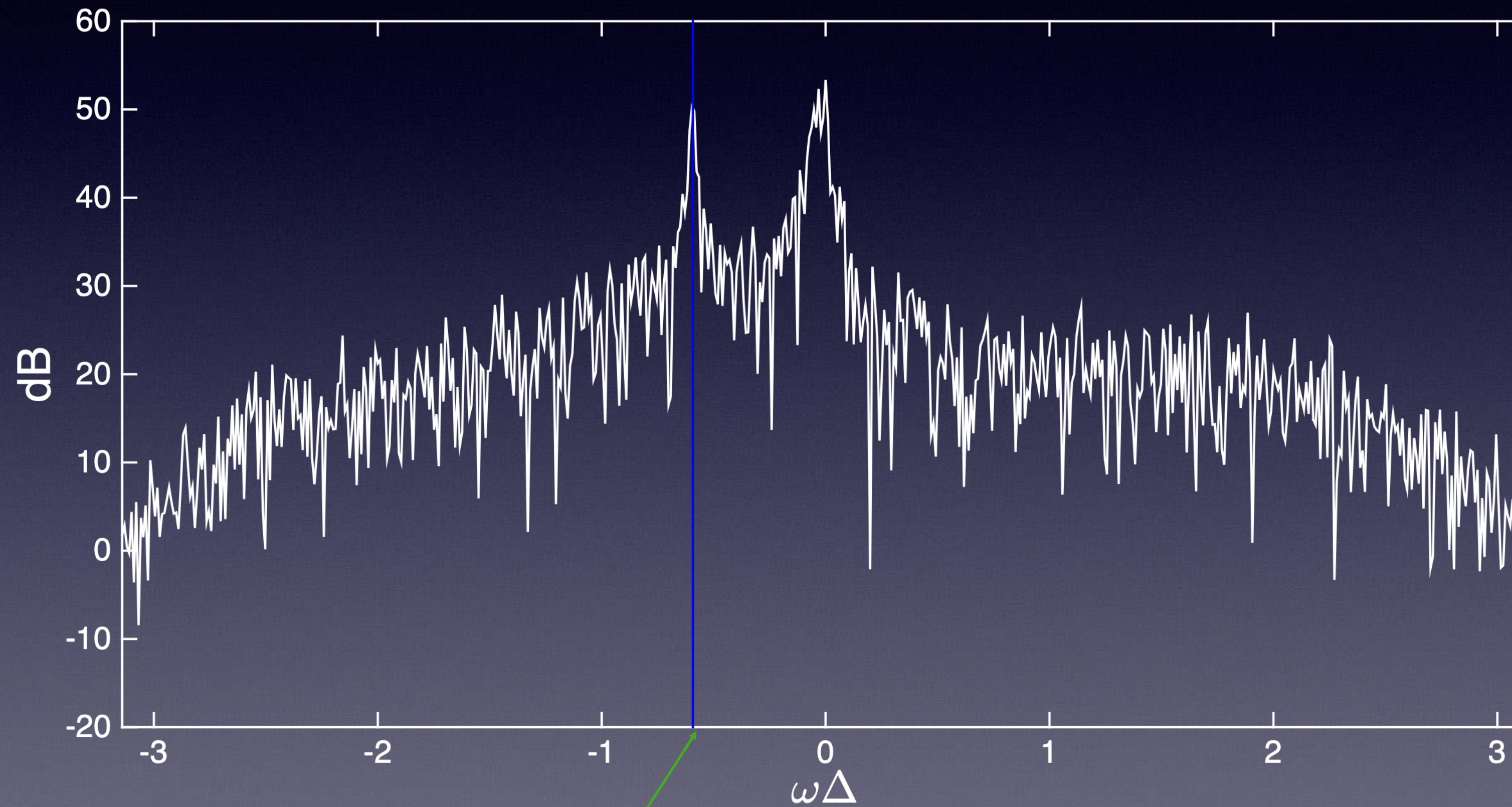
Represent particle velocities as complex-valued time series:

$$z_t = u_t + iv_t$$

Power spectral density for velocities from:



Power spectral density for velocities from:



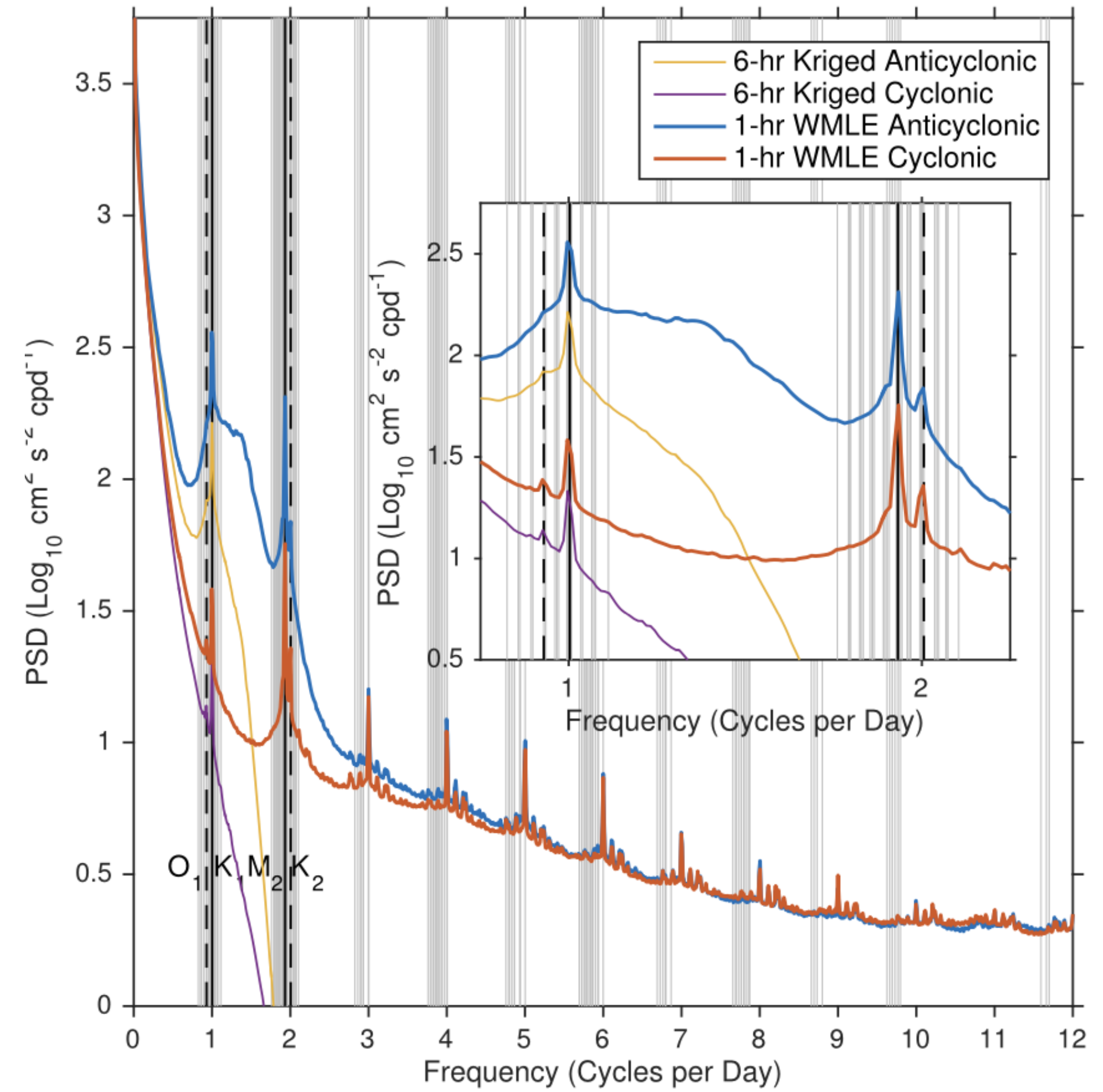
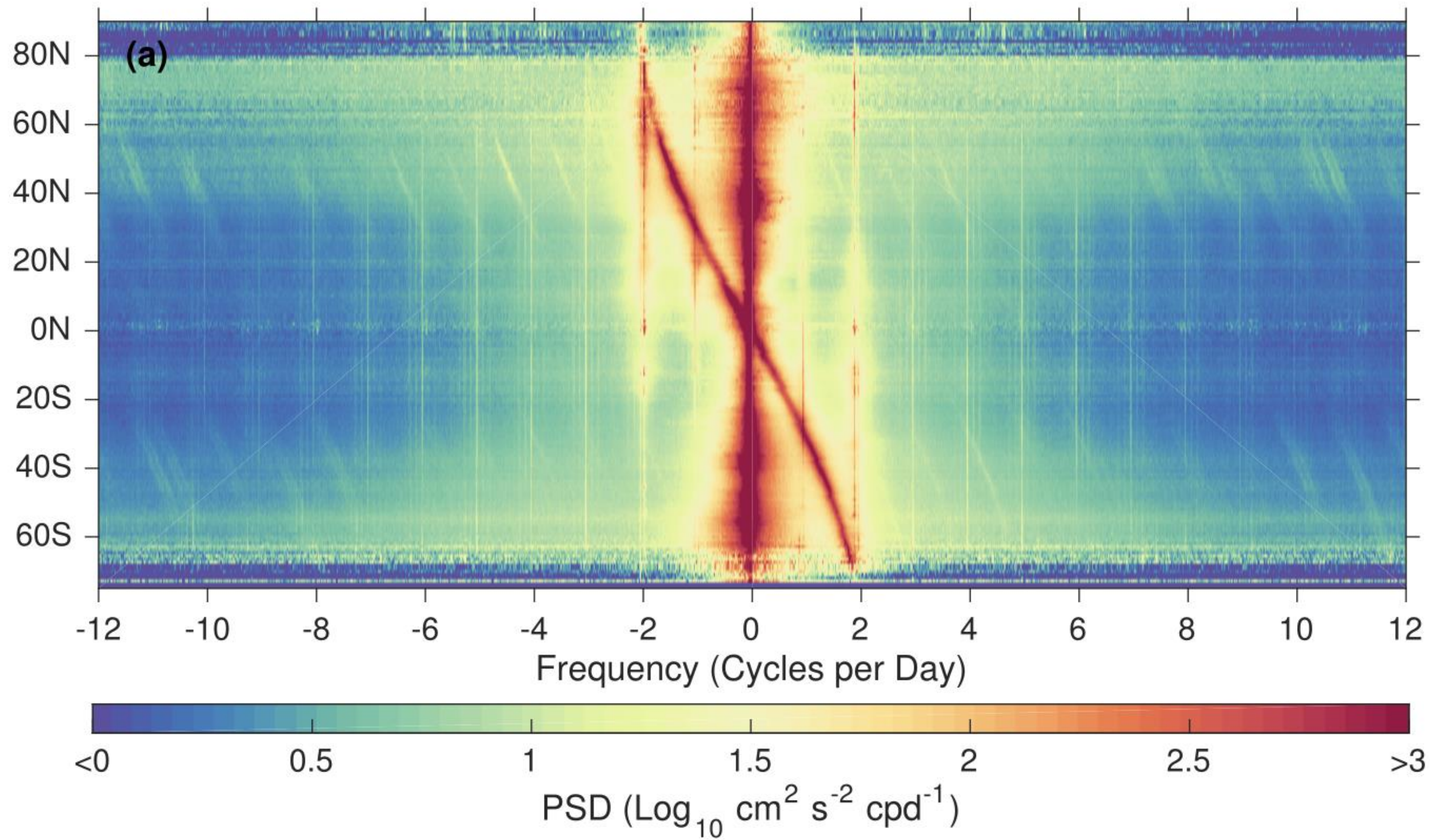
Coriolis frequency

Nonstationarity: Time-frequency “spectrograms” and Heisenberg-Gabor uncertainty



Time in days

Peaks in the power spectral density...



Head over to RStudio for Activity 2!